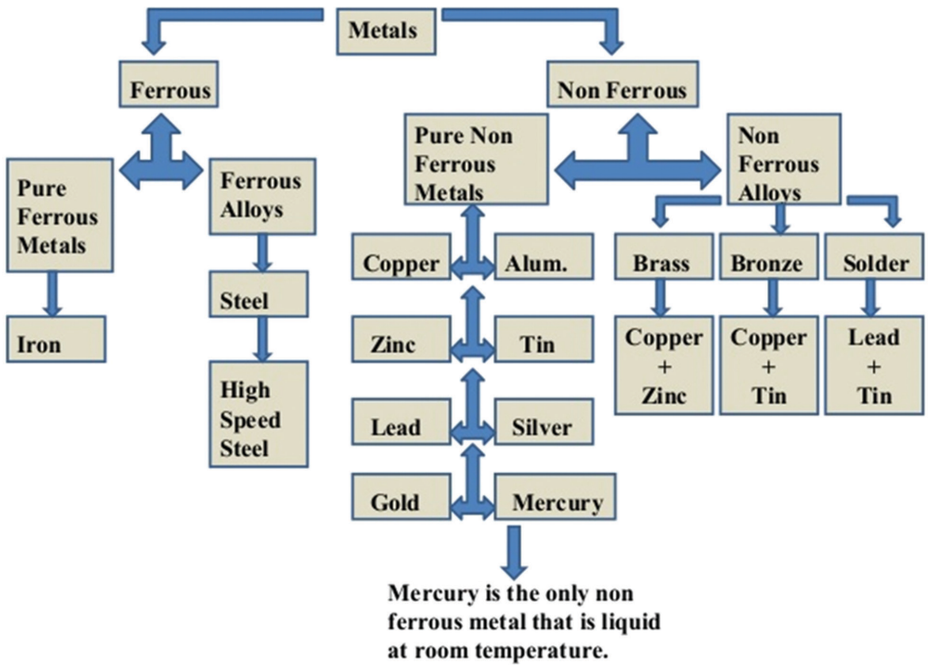


Fig 9.3 *Classification of Metals*

- Indians have specialized in metallurgy of iron and steel manufacturing.
- Ashoka's pillar at Delhi, Puri temple's iron joints speak the glory of our skill.

9.2.3 NON- FERROUS METAL

- The metal which contains metals other than iron as main constituent is non-ferrous metal.
- Copper, Aluminium, Zinc, Lead, Tin, Platinum, Silver, Gold, Tungsten are some examples of non- ferrous metals.

9.2.4 ALLOY METAL

- Combining of two or more metals to obtain special properties are called as alloy metal.
- Brass, Bronze, Stainless steel, Nichrome are some example of alloy metal.

9.3 CLASSIFICATION OF MATERIALS AND PROPERTIES

Properties of materials can be classified under several heads. As a matter of fact the following, classification of materials are important for an engineer, to select a suitable material for a desired purpose.

9.3.1 PHYSICAL PROPERTIES

Physical properties determine the micro and macro structure of the materials such as shape, size, colour, lustre , density , structure, finish , etc..

9.3.2 MECHANICAL PROPERTIES

These properties deals with behaviour of materials while force or load is acting on it, such as elasticity, plasticity, ductility,

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